Strategic renewal in the age of artificial intelligence

Received (in revised form): 9th February, 2025

Jacques Bughin
Chief Executive Officer, Machaon Advisory, Belgium

Dr Jacques Bughin is currently Chief Executive Officer of Machaon Advisory, specialising in digital technology economics and top management consultancy. Previously, he was a senior partner at McKinsey & Co. and leader of the McKinsey Global Institute. He collaborates with multiple institutions, from the United Nations (UN), the Science and Technology Options Assessment (STOA) Board of the European Parliament to the Portulans Institute and the Accenture Research Board. He has been a professor of management practices and is now a senior adviser to private equity companies Fortino Capital and Antler. He has published widely on technology and artificial intelligence (AI) in, among others, *Sloan Management Review*, *Harvard Business Review*, *Technovation*, *Frontiers in Artificial Intelligence* and *European Business Review*.

Machaon Advisory, Avenue Montana 24 1180, Bruxelles, Belgium
E-mail: bughinjacquesrenejean@gmail.com; jacquesbughin@machaonadvisory.com

Abstract Strategic renewal is essential for companies seeking to maintain a competitive edge in an era of rapid technological disruption, such as artificial intelligence (AI). Our findings reveal that while 45 per cent of companies acknowledge the need for renewal, only 17 per cent pursue radical transformation. Barriers such as AI capability gaps, leadership misalignment and the disruption dilemma hinder effective adaptation. Companies that successfully integrate AI-driven strategic renewal, however, demonstrate a balanced approach between exploration and exploitation, proactive investment in AI capabilities and strong leadership commitment. Through case studies of companies including IBM, NVIDIA and JPMorgan, this paper highlights key drivers of AI-led transformation and offers actionable strategies for executives. Ultimately, this research underscores that the future belongs to incumbents who proactively reinvent themselves in the AI age. This article is also included in **The Business & Management Collection** which can be accessed at https://hstalks.com/business/.

KEYWORDS: strategic renewal, AI disruption, business transformation, organisational agility, leadership, digital reinvention, competitive advantage

DOI: 10.69554/CZRB1854

WHY STRATEGIC CHANGE?

There is never a good time to reinvent, yet a unique defining capability of a company is to know when and how to change.[1] The art of reinvention is usually called by strategy theorists 'strategic renewal', and is defined as an organisation's proactive approach to adapting and reshaping its strategy to maintain competitive advantage in response to major external disruptions.[2,3]

The reason why strategy theorists have dedicated time to studying strategy renewal is not for fun,[4] but because they have realised that in turbulent environments, typical market structure and conduct are unstable to warrant sustained performance, and that a strategic renewal is often a defining action for an organisation's future. Multiple research suggests that high-performing companies reinvent themselves every five to ten years, while those that fail to do so risk stagnation and decline.[5] Other business analysts have claimed that more than 40 per cent of companies had to reinvent every three years to survive.[6]

The issue of being too late is as acute as the issue of not reinventing. Nokia, once a global leader in mobile phones, faced near collapse after failing to transition quickly enough to the smartphone era. Yet 'failure can be a trigger for sense-making efforts and a rich information source for learning';[7] with new insights, Nokia strategically pivoted towards telecommunications infrastructure and technology services, securing multi-billion-dollar contracts and achieving profitability again, and strong momentum, by the start of 2025.[8]

Kodak presents another example of a company that struggled with change but is still prevailing, although as a very different company than it was pre-digital times.[9] It is widely known that despite inventing the digital camera, Kodak failed to shift its business model in time, leading to its bankruptcy in 2012. Since then, however, the company has diversified into commercial printing, speciality chemicals and imaging technologies in an attempt to regain relevance. More recently, Kodak has modernised its film manufacturing plant to meet renewed demand for analogue photography, signalling another step toward strategic renewal.[10,11]

But companies do not all wait 20 (Nokia) or 40 (Kodak) years to thrive again. IBM, for example, has continuously evolved through multiple reinventions. Founded in 1911 as a tabulating machine company, it has successfully transformed itself multiple times — from mainframes to personal computers, to cloud computing, and now artificial intelligence (AI). IBM's ability to anticipate change and restructure its operations has allowed it to remain relevant in technology for over a century. Similarly, Samsung, once a trading company in the 1930s, transitioned into electronics in the 1960s and later became a leader in semiconductors, mobile technology and AI-powered innovations. Samsung's capacity to continuously invest in research and development (R&D), diversify its portfolio and integrate AI-driven solutions has kept it ahead of competitors such as Toshiba which failed to pivot in time.

A UNIQUE DISRUPTOR CRYING FOR CHANGE: AI

The necessity of strategic renewal extends beyond specific industries and regions. The accelerating pace of technological advancements demands that companies evolve continually. In fact, studies clearly indicate that companies have more and more trouble surviving; the average life span of an S&P 500 company has shrunk from 65 years in 1947 to just 18 years today.[12,13]

Ultimately, success in strategic renewal does not happen by chance. With the spread of AI fundamentally reshaping industries by driving automation and innovation, there is a unique opportunity to study how companies are proactively operating strategic changes. For many, AI currently represents the most profound disruptive force, as it encompasses all factors leading to disruption.[14] In effect, disruptive technology theories emphasise how disruption emerges from technology substitution,[15] revolutionary innovations[16] and killer technologies.[17]

Technology substitution occurs when a new technology gradually replaces an existing one. AI is currently undergoing this process in numerous fields, automating and replacing human expertise such as AI-driven radiology or drug discovery in healthcare.[18]

Revolutionary innovations do not just replace old technologies but redefine industries and value chains by altering how businesses operate. AI-driven large language models (LLMs) and generative AI (GenAI) are doing precisely this in capital markets. The well-known hedge fund company Renaissance Technologies has introduced deep machine learning (ML) techniques and LLM tools into its own algorithmic trading, seamlessly blending quantitative finance and AI as the new playing field in capital markets.[19]

Finally, killer technologies are those that completely wipe out incumbent companies or industries. AI-powered robotic process automation (RPA) by companies such as UiPath and Automation Anywhere are deploying AI-driven RPA solutions that have the potential to eliminate entire back-office functions in industries such as finance, insurance and customer service. AI is also seen as the 'killer technology' behind the decline of traditional advertising agencies.[20] AI-powered ad personalisation is erasing the market dominance of ad agencies that once controlled brand strategy and media placement. GenAI is also killing the creative arm of ad agencies as campaigns can be crafted much faster, while new AI visual tools reduce reliance on traditional graphic resources at creative agencies.[21]

THE REINVENTION IN ACTION FOR AI

We recently studied a random sample of 600 large companies worldwide and checked on their strategic renewal actions in the context of AI transformation. The sample is pre-deployment of GenAI.

As expected, AI stress — defined as competitive pressure from AI-driven companies — emerges as a key catalyst for strategic renewal. This is good news as it means that chief executive officers (CEO) and managers are conscious of the potential destructive effects of AI.

Despite growing awareness of AI's impact, however, most companies continue to struggle with full change. Our findings indicate that while 45 per cent of surveyed companies want to engage in strategic renewal in response to AI disruption, only 17 per cent have been pursuing radical changes. This suggests that many organisations are failing to fully capitalise on the transformative potential of AI and remain stuck in incremental change rather than fundamental reinvention.[22]

Finally, even for those companies that opt for radical changes, success is not easy, as observed in the past.[23] Those that thrive are the ones that proactively invest in AI capabilities (skill, data and ML) rather than merely reacting to external pressures. These companies also integrate AI across both automation and product innovation to maintain competitive advantage and ensure strong leadership involvement in AI-driven transformation.

WHY INCUMBENTS STRUGGLE WITH AI DISRUPTION

While AI presents new opportunities, many large companies face three significant barriers that hinder their proactive renewal. First, there is the disruption dilemma, where companies must decide whether to fully embrace AI — potentially cannibalising existing revenue streams — or maintain traditional models and risk gradual market erosion. By going too quickly, early versions of AI technologies may be bypassed by more improved features, as currently seen in the race of improvements in LLMs.[24]

Timing, however, is a choice[25] and proactivity should not be done with a sense of industry comparison, as the disruption is often external to the industry. This proactive agility entails uncertainty and also requires to have enough resource slack to allow for adjustment.

Capability gaps continue to hinder progress, with 40 per cent of companies citing data accessibility as a primary barrier to AI-driven renewal. Additionally, only 20 per cent of companies report having a workforce with sufficient skills to lead AI initiatives.[26] Finally, there is a clear lack of strategic alignment at the leadership level, as only 23 per cent of companies report having CEO-led AI strategies. This reveals a significant gap between AI awareness and execution, which limits the effectiveness of transformation efforts.[27]

KEY DRIVERS OF AI-DRIVEN STRATEGIC RENEWAL

Successful AI-driven strategic renewal is determined by four primary factors. The first is external players signalling AI stress, which acts as a catalyst for transformation. SAP responded to rising AI competition from cloud-native ERP providers such as Workday by integrating AI-powered business process automation. As a result, SAP increased its cloud revenue by 40 per cent in 2023, demonstrating the importance of proactive AI investment.[28]

The second key factor is leadership commitment. Companies with CEOs who actively engage in AI strategy are significantly more likely to undergo full-scale AI-driven renewal, with studies showing a 6 per cent greater likelihood of transformation. JPMorgan CEO Jamie Dimon personally championed the company's AI-driven fraud detection and investment algorithms, giving the company a significant data-driven advantage in banking.[29] Philips' CEO Roy Jakobs was behind steering the company through a significant transformation, leveraging AI to enhance medical diagnostics and patient care.[30] CEO Steve Hasker has led the charge at Thompson Reuters to move to an AI-first organisation,[31] incorporating GenAI into its flagship products. The company is quickly enlarging its ecosystem, acquiring Casetext, a provider of AI-driven legal technology, and entering into licensing agreements with AI companies such as Meta.[32]

The third factor is balancing exploration and exploitation. Firms that strike a balance between using AI for cost-reduction automation and for launching new AI-driven products and services see a larger return on AI investments. Pfizer, despite the absence of obesity drugs in its portfolio that have boosted the stock price of the likes of Novo Nrodisk and Eli Lilly, has beaten its peers in sales growth lately;[33] one reason is the aggressive play in AI of the company to expedite drug discovery, optimised clinical trials and enhanced manufacturing processes, which started with the collaboration with BioNTech of the COVID-19 vaccine.[34]

Another essential driver of renewal is data accessibility and AI capabilities. Companies that do not face data accessibility constraints are 9 per cent more likely to engage in strategic renewal. European car manufacturer Volvo has successfully leveraged AI-powered safety and navigation systems, positioning itself ahead of competitors that continue to struggle with data integration and AI model training.

STRATEGIC RENEWAL PAY OFF

The above clearly demonstrates large heterogeneity in strategic renewal actions among firms. These results can be compared with another study, this time performed on a different set of global firms in late 2021.[35] Besides typical AI investments, there is clear evidence that strategy renewal is the largest component of the impact of AI on the bottom line of companies — just above innovation and leadership. Those effects are robust, based on ML techniques, that control for all interactions between AI assets, capabilities and factors such as strategy renewal.[36]

WHAT EXECUTIVES MUST DO NOW

As strategy renewal is a key ingredient of AI transformation, proactive CEOs should be motivated to undertake the following actions sooner rather than later.

Action 1:

- *Evaluate current profit streams*: Only 17 per cent of companies are pursuing radical AI-driven change, while 45 per cent acknowledge the need for renewal.
- *Identify external threats*: AI disrupts

industries through technology substitution (eg radiology, drug discovery), revolutionary innovations (eg LLMs in capital markets) and killer technologies (eg RPA eliminating back-office functions).
- *Explore new market opportunities*: AI opens new revenue streams; most successful companies that have changed business models have diversified and entered new AI ecosystems.

Action 2: Integrate your AI renewal with the following complementary actions:

- Foster a culture of AI experimentation and innovation.
- *Encourage AI-led innovation*: Renaissance Technologies introduced deep learning (DL) into hedge fund trading, blending quantitative finance and AI.
- *Embrace external AI*: Microsoft integrated OpenAI's ChatGPT into enterprise software (Copilot, Azure AI) instead of resisting disruption.
- Shape the right and align leadership.
- *Ensure leadership buy-in*: JPMorgan CEO Jamie Dimon championed AI-powered fraud detection and trading algorithms, giving the bank a data-driven competitive advantage.
- *Communicate AI strategy clearly*: Samsung leadership pushed its organisation systemically on AI until it successfully integrated AI into camera and smartphone assistants, helping it maintain a premium market position.
- *Mandate AI training for your C-suite*: Mandatory training for C-suite has twice the impact on AI scaling success than training of employees.
- Aggressively develop new complementary capabilities (talent and data).
- *Invest in organisation AI upskilling*: IBM shifted from traditional computing to AI-driven enterprise solutions, training thousands of employees in AI.
- Acquire AI talent who can shortcut time from training to actions; build AI influencers; establish AI ecosystems to limit bottlenecks.

Action 3: Balance AI exploration versus exploitation:

- Strategic renewal for incumbents is a balancing act of exploration and exploitation, as what differentiates an incumbent from a new company is the incumbent's ability to leverage its existing strengths and resources.

NVIDIA may exemplify this balanced approach. On one hand, NVIDIA has dominated the graphics processing unit (GPU) market, catering to gamers, professional designers and data centres. It has leveraged its core GPU technology to expand into AI acceleration, cloud computing and high-performance computing. The company continues to refine its RTX series of GPUs, focusing on improving efficiency, speed and AI-driven performance for existing customers. On the other hand, NVIDIA has successfully reinvented itself by expanding into AI, autonomous vehicles and data centre computing. Investments in DL and AI chips (such as the NVIDIA H100 and A100 Tensor Core GPUs) have positioned the company as a leader in AI acceleration.

The outcome of a good balance is the evidence of major synergy. NVIDIA does not abandon its core GPU business but integrates AI and DL into its existing technologies (eg deep learning super sampling [DLSS] improves gaming performance using AI).

POSTSCRIPT

The future belongs to AI-driven incumbents who make strategic renewal today. Are you/your company one of them?

References and Notes

1. Bertolini, M., Duncan, D. S. and Waldeck, A. (December 2015), 'Knowing When to Reinvent', Harvard Book Review, available at https://hbr.

org/2015/12/knowing-when-to-reinvent (accessed 9th February, 2025).

2. Bughin, J. and Van Zeebroeck, N. (2017), 'The best response to digital disruption', *MIT Sloan Management Review*, Vol. 58, No. 4, pp. 80–86.
3. Ciszewska-Mlinarič, M. and Wójcik, P. (2023), 'What do we talk about when we talk about strategic renewal: A systematic literature review of 40 years of research', *Central European Management Journal*, Vol. 31, No. 3, pp. 416–441.
4. Napolitano, M. R., Marino, V. and Ojala, J. (2015), 'In search of an integrated framework of business longevity', *Business History*, Vol. 57, No. 7, pp. 955–969.
5. Bhalla, V., Caye, J-M., Dyer, A., Dymond, L., Morieux, Y. and Orlander, P. (September 2011), 'High-Performance Organizations', Boston Consulting Group (BCG), in *Own the Future: 50 Ways to Win from the Boston Consulting Group*, pp. 171–177.
6. Reinvention Academy (May 2020), 'How Often Should You Reinvent to Survive?', available at https://www.learn2reinvent.com/blog/how_often_should_you_reinvent_to_survive (accessed 9th February, 2025).
7. Byrne, O. and Shepherd, D. A. (2015), *Different Strokes for Different Folks: Entrepreneurial Narratives of Emotion, Cognition, and Making Sense of Business Failure*, Wiley, Hoboken, NY, p. 375.
8. Wood, N. (January 2025), 'Nokia ended 2024 on a high note, as net sales finally turn the corner', Telecoms, available at https://www.telecoms.com/operator-ecosystem/nokia-ended-2024-on-a-high-note-as-net-sales-finally-turn-the-corner (accessed 9th February, 2025).
9. Made-in-China (November 2024), 'Does the Kodak company exist today?', available at https://insights.made-in-china.com/Does-the-Kodak-company-still-exist_ptATDvZGqQHe.html#:~:text=Today%2C (accessed 9th February, 2025).
10. Vinokurova, N. and Kapoor, R. (August 2023), 'Kodak's surprisingly long journey towards strategic renewal: A half-century of exploring digital transformation in the face of uncertainty and inertia', *Academy of Management Proceedings*, Vol. 1.
11. Mukherjee, A. (January 2025), 'Nokia beats estimates as demand recovers, shares rise', Reuters, available at https://www.reuters.com/business/media-telecom/nokias-q4-profit-beats-estimates-2025-01-30/?utm_source=chatgpt.com (accessed 9th February, 2025).
12. Garelli, S. (December 2016), 'Why you will probably live longer than most big companies', International Institute for Management Development (IMD), available at https://www.imd.org/research-knowledge/disruption/articles/why-you-will-probably-live-longer-than-most-big-companies/#%3A%7E%3Atext%3DA%20recent%20study%20by%20McKinsey%20found%20that%20the%2Cquoted%20on%20the%20S%26P%20500%20will%20have%20disappeared (accessed 9th February, 2025).
13. Viguerie, S. P., Calder, N. and Hindo, B. (May 2021), '2021 Corporate Longevity Forecast', Innosight, available at https://www.innosight.com/insight/creative-destruction/ (accessed 9th February, 2025).
14. Singh, N., Chaudhary, V., Singh, N., Soni, N. and Kapoor, A. (2024), 'Transforming business with generative AI: Research, innovation, market deployment and future shifts in business models', arXiv, available at https://arxiv.org/pdf/2411.14437 (accessed 9th February, 2025).
15. Fisher, J. C. and Pry, R. H. (1971), 'A simple substitution model of technological change', *Technology Forecasting and Social Change*, Vol. 3, Nos. 2–3, pp. 75–88.
16. Abernathy, W. J. and Clark, K. B. (1985), 'Innovation: Mapping the winds of creative destruction', *Research Policy*, Vol. 14, No. 1, pp. 3–22.
17. Coccia, M. (2019), 'Killer technologies: The destructive creation in the technical change', arXiv, available at https://arxiv.org/pdf/1907.12406 (accessed 9th February, 2025).
18. Jayatunga, M. K. P., Ayers, M., Bruens, L., Jayanth, D. and Meier, C. (2024), 'How successful are AI-discovered drugs in clinical trials? A first analysis and emerging lessons', *Drug Discovery Today*, 104009.
19. S, K. (October 2023), 'Decoding the Secrets of Renaissance Technologies: The Machine Learning Magic Behind Jim Simons' Success', IBM, available at https://community.ibm.com/community/user/ai-datascience/blogs/kiruthika-s2/2023/10/23/decoding-the-secrets-of-renaissance-technologies (accessed 9th February, 2025).
20. Schwarz, R. (November 2024), 'How Generative AI Is Disrupting Today's Advertising Agency Model', *Forbes*, available at https://www.forbes.com/councils/forbescommunicationscouncil/2024/11/25/how-generative-ai-is-disrupting-todays-advertising-agency-model/?utm_source=chatgpt.com (accessed 9th February, 2025).
21. Agarwal, T., Gopalkrishnan, S., Kale, V., Periwal, D., Kulkarni, A. A., Tharkude, D. and Jaiwani, M. (November 2023), 'Transforming advertising: Harnessing AI for personalised customer-centricity', 2023 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD), pp. 1–8.
22. Ciszewska-Mlinarič and Wójcik, ref. 3 above.
23. Uotila, J., Maula, M., Keil, T. and Zahra, S. A. (2009), 'Exploration, exploitation, and financial performance: Analysis of S&P 500 corporations', *Strategic Management Journal*, Vol. 30, No. 2, pp. 221–231.
24. Bughin, J. (2024), 'Inside the successful make-up of "AI-first" organisations', *Journal of AI, Robotics & Workplace Automation*, Vol. 3, No. 3, pp. 16–25.
25. Volberda, H. W., Van den Bosch, F. A., Flier, B. and Gedajlovic, E. R. (2001), 'Following the herd or not?: Patterns of renewal in the Netherlands and the UK', *Long Range Planning*, Vol. 34, No. 2, pp. 209–229.
26. Bughin, ref. 24 above.
27. Mikalef, P. and Gupta, M. (2021), 'Artificial intelligence capability: Conceptualization,

measurement calibration, and empirical study on its impact on organizational creativity and firm performance', *Information & Management*, Vol. 58, No. 3, 103434.
28. SAP News (January 2025), 'SAP announces Q4 and FY 2024 Results', press release, available at https://news.sap.com/2025/01/sap-announces-q4-and-fy-2024-results/ (accessed 9th February, 2025).
29. JPMorgan (2023), 'JPMorgan CEO Jamie Dimon: AI Could "Augument Virtually Every Job"', available at https://www.pymnts.com/news/artificial-intelligence/2024/jpmorgan-ceo-jamie-dimon-ai-could-augment-virtually-every-job/ (accessed 9th February, 2024).
30. Patel, N. (September 2024), 'How Philips CEO Roy Jakobs is turning the company around after a major recall', The Verge, available at https://www.theverge.com/24242833/philips-ceo-roy-jakobs-ai-healthcare-fda-cpap-recall-decoder-podcast-interview?utm_source=chatgpt.com (9th February, 2025).
31. Bughin, ref. 24 above.
32. Thomson Reuters (November 2023), 'Thomson Reuters unveils generative AI strategy designed to transform the future of professionals', available at https://www.thomsonreuters.com/en/press-releases/2023/november/thomson-reuters-unveils-generative-ai-strategy-designed-to-transform-the-future-of-professionals?utm_source=chatgpt.com (accessed 9th February, 2025).
33. Dunleavy, K. (November 2024), 'Pfizer led industrywide sales surge in Q3 after several quarters of dominance by Eli Lilly, Novo Nordisk', Fierce Pharma, available at https://www.fiercepharma.com/pharma/while-lilly-and-novos-momentum-stalled-q3-pfizer-moved-top-spot-leading-industry-wide-sales?utm_source=chatgpt.com (accessed 9th February, 2025).
34. Silver, K., 'Shot of a Lifetime: How Pfizer and BioNTech Developed and Manufactured a COVID-19 Vaccine in Record Time', Pfizer, available at https://www.pfizer.com/news/articles/shot_of_a_lifetime_how_pfizer_and_biontech_developed_and_manufactured_a_covid_19_vaccine_in_record_time?utm_source=chatgpt.com (accessed 9th February, 2024).
35. Bughin, ref. 24 above.
36. Results are based on Lasso regression techniques, with optimal penalties derived from the data. Significance of factors tested by run of bootstrap.

APPENDIX: SAMPLE HIGHLIGHT

Data originates from a selected sample of organisations from a large-scale survey conducted by TNS Sofres for McKinsey & Company on digital business transformation across industries. The sample covers 545 companies cross-validating survey answers with external Capital IQ data. Final sample includes companies from seven key sectors including automotive, financial services, healthcare, high-tech, media, retail and business services. Thirty-eight per cent are US/Canada, 20 per cent top 5 European countries and 18 per cent China. Seven per cent response rate. Multiple bias control methods applied, ML validation and common factor analysis confirmed reliability.

